# Md Mamunur Rashid

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## SUMMARY OF QUALIFICATIONS

- PhD candidate at Deakin University researching trustworthy AI, federated learning, adversarial robustness, and privacy.
- 9.5 years of telecommunications experience in operations, analytics, automation, and technical management.
- Five published journal papers, five published conference papers, and a paper accepted at NeurIPS 2026; additional manuscripts under review.
- Deakin Graduate Research and Teaching Fellow supporting computer networks, secure coding, malware analysis, and cloud computing.
- Proficient in Python, PyTorch, TensorFlow, C/C++, SQL, and blockchain development (Solidity/Ethereum/Hyperledger).
- Certified Data Analyst (ML), Certified ScrumMaster (CSM), and CCNA-certified professional.

## EDUCATION

- **Doctor of Philosophy (PhD)**, Information Technology — ***Nov 2023 – Present***
  Deakin University, Victoria, Australia
  *Thesis: Enhancing Security and Trust in Federated Learning*
  Thesis submission planned for 7 November 2026; examination and degree conferral pending.
- **Master of Engineering (MEng)**, AI Convergence — ***Mar 2021 – Feb 2023***
  Pukyong National University (PKNU), Busan, South Korea — CGPA: 4.45/4.50
  *Thesis: A Novel Intrusion Detection System in IoT Networks Leveraging Blockchain-Enabled Federated Learning*
- **Bachelor of Science (BSc)**, Electronics & Telecommunication Engineering — ***Feb 2007 – Jun 2011***
  RUET, Bangladesh — CGPA: 3.56/4.00

## ACADEMIC & TEACHING EXPERIENCE

**Deakin University, School of Information Technology, Melbourne, Australia** — ***Nov 2023 – Present***
*Graduate Research & Teaching Fellow*

- Graduate Research & Teaching Fellow for SIT202 Computer Networks and Communication, SIT324 Malware Analysis, SIT218 Secure Coding, and SIT233/SIT706 Cloud Computing.
- Facilitate seminars, support students, and mark assessments in security, networking, and cloud computing units.

**Pukyong National University (PKNU), MCSP Lab, Busan, South Korea** — ***Mar 2021 – Sep 2023***
*Teaching & Research Assistant*

- Instructed graduate courses: Introduction to ML/DL, Cybersecurity Analytics, Computer Networks, and Data Communications.
- Supervised junior lab members and presented research outcomes at inter-university collaborative seminars.

## INDUSTRY EXPERIENCE

**Robi Axiata Limited (Multinational Telecom Operator), Dhaka, Bangladesh** — ***Apr 2020 – Mar 2021***
*Manager, Technology Operations*

- Governed outage and trouble ticket workflows across Huawei, Ericsson, and Nokia ecosystems.
- Analysed and implemented AI/RPA solutions for forecasting network capacity and incidents.

**Robi Axiata Limited, Dhaka, Bangladesh** — ***Jun 2019 – Mar 2020***
*Specialist, Analytics & Reporting*

- Developed network performance dashboards and fault reports distributed to senior management.

**Robi Axiata Limited, Dhaka, Bangladesh** — ***Jul 2017 – May 2019***
*Shift Manager, Service Operation Center*

- Led 24/7 operational shifts ensuring business continuity across nationwide 2G/3G/4G networks.

**Robi Axiata Limited, Dhaka, Bangladesh** — ***Nov 2011 – Jun 2017***
*Specialist, First Line Support*

- Provided first-line support across Radio, Core, and IP domains. Monitored KPIs and diagnosed system faults.

## PUBLICATIONS

*Published Journal Articles (5)*

[J1] **M. M. Rashid**, Y. Xiang, M. P. Uddin, J. Tang, K. Sood, and L. Gao, "Trustworthy and Fair Federated Learning via Reputation-Based Consensus and Adaptive Incentives," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 2868–2882, 2025. doi:10.1109/TIFS.2025.3546841.

[J2] **M. M. Rashid**, M. P. Uddin, Y. Xiang, K. Sood, Y. Zhang, and L. Gao, "TAFL: Trust-Aware Federated Learning for Anomalous Client Detection, Isolation, and Recovery," *IEEE Transactions on Information Forensics and Security*, vol. 21, pp. 9202–9216, 2026. doi:10.1109/TIFS.2026.3744607.

[J3] **M. M. Rashid**, S. U. Khan, F. Eusufzai, M. A. Redwan, S. R. Sabuj, and M. Elsharief, "A Federated Learning-Based Approach for Improving Intrusion Detection in Industrial Internet of Things Networks," *Network*, vol. 3, no. 1, pp. 158–179, 2023. doi:10.3390/network3010008.

[J4] **M. M. Rashid**, P. Choi, S.-H. Lee, and K.-R. Kwon, "Block-HPCT: Blockchain Enabled Digital Health Passports and Contact Tracing of Infectious Diseases like COVID-19," *Sensors*, vol. 22, no. 11, art. 4256, 2022. doi:10.3390/s22114256.

[J5] **M. M. Rashid**, S.-H. Lee, and K.-R. Kwon, "Blockchain Technology for Combating Deepfake and Protect Video/Image Integrity," *Journal of Korea Multimedia Society*, vol. 24, no. 8, pp. 1044–1058, 2021. doi:10.9717/kmms.2021.24.8.1044.

*Conference Papers (1 accepted; 5 published)*

[C1] **M. M. Rashid**, Y. Xiang, M. P. Uddin, *et al.*, "Privacy Risk Scales with Effective Dimension in Federated Learning," *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. **Accepted; proceedings publication forthcoming.**

[C2] **M. M. Rashid** *et al.*, "Ensuring Privacy and Security of IoT Networks Utilizing Blockchain and Federated Learning," *10th International Conference on Future Internet of Things and Cloud (FiCloud)*, pp. 298–305, 2023.

[C3] **M. M. Rashid** *et al.*, "A Blockchain-Based Approach in Healthcare Supply Chain Using Smart Contracts and Decentralized Storage Systems," *ACM Conference on Information Technology for Social Good (GoodIT)*, 2022. doi:10.1145/3524458.3547251.

[C4] **M. M. Rashid** *et al.*, "Utilizing Blockchain and Distributed Storage to Enhance Security and Privacy in the IoT Ecosystem," *IEEE Global Conference on Artificial Intelligence and Internet of Things (GCAIoT)*, 2022. doi:10.1109/GCAIOT57150.2022.10019111.

[C5] **M. M. Rashid** *et al.*, "Emergence of the Metaverse: Challenges and Opportunities," *International Conference on Multimedia Information Technology and Applications (MITA)*, 2022.

[C6] **M. M. Rashid** *et al.*, "A Conceptual Blockchain Framework for Protection of Multimedia Content," *Korea Multimedia Society Fall Conference*, 2021.

*Selected Manuscripts Under Review / In Revision (3)*

[M1] **M. M. Rashid** *et al.*, "ZK-TrustChains: Unlinkable Reputation via Recursive Zero-Knowledge Proofs for Sybil-Resistant Federated Learning," *IEEE Transactions on Reliability*, revised manuscript under review.

[M2] **M. M. Rashid** *et al.*, "Beyond Round-Local Aggregation: Learning Cross-Session Client Trajectories for Robust Federated Learning," *The Web Conference 2027*, under review.

[M3] **M. M. Rashid** *et al.*, "Soul-FL: Mitigating Sleeper Sybil Attacks in Federated Learning via Statistically Anchored Identities and Decaying Trust," *The Web Conference 2027*, under review.

## RESEARCH PROJECTS

- **Trust, Security, and Privacy in Federated Learning**, Deakin University, Australia ***2023 – Present***
  Developed reputation-based consensus and incentives, trust-aware anomalous-client detection and recovery, and privacy-risk analyses for distributed AI; reported in IEEE TIFS (2025, 2026) and NeurIPS 2026 (accepted).
- **Deepfake Detection using Deep Learning**, PKNU, Busan, South Korea ***2021 – 2022***
  Implemented CNN, LSTM, GRU, and Vision Transformer-based detection algorithms using PyTorch/Python. Published in J. Korea Multimedia Society.
- **Multimedia Content Protection using Blockchain & AI**, PKNU, Busan, South Korea ***2021 – 2022***
  Converged blockchain and deep learning to verify and protect image/video integrity. Published in KMMS 2021.
- **Incident Capture & AI/RPA Automation Tools**, Robi Axiata Limited, Dhaka, Bangladesh ***2019 – 2020***
  Developed AI-driven incident capture tool correlating cross-domain alarms, and an RPA tool for site downtime notifications. Deployed in production.

## ACADEMIC SERVICE & MEMBERSHIPS

- **Invited reviewing and program-related service:** USENIX Security 2027; IEEE TIFS, TDSC, TMC, TCC, IoT-J, and TKDE; ACM CSUR and TOPS.
- **Member:** IEEE (Graduate Member).

## TALKS & PRESENTATIONS

- **3-Minute Thesis (3MT) Competition**, School of Information Technology, Deakin University ***Jun 2025***
  *Talk: "Anchoring Identity and Tracing Deception: Toward Trustworthy Federated Learning"*
- **HDR Conference**, School of Information Technology, Deakin University ***Nov 2025***
  *Talk: "Toward Trustworthy and Secure Federated Learning"*
- **Guest Speaker**, Pukyong National University (PKNU), Busan, South Korea ***Aug 2022***
  *Talk: "Blockchain Technology for Secure and Transparent Healthcare Supply Chain Management"*

## HONORS & AWARDS

- **Best Paper Award** by Korea Multimedia Society (KMMS) Fall Conference, 2021
- **Champion — "Innovate to Inspire"** Guiding Principle Award by Robi Axiata Limited, Q3 2019
- **Robi Accelerated Development Program (RADP)** — Selected member, internal high-potential leadership programme
- **Board Merit Scholarship** by Government of Bangladesh, Secondary and Higher Secondary Examinations

## CERTIFICATIONS

- **Certified Data Analyst (Machine Learning)** — Axiata / Coursera (University of Michigan & UC Davis)
- **Certified ScrumMaster (CSM)** — Scrum Alliance
- **Cisco Certified Network Associate (CCNA)** — RUET, 2009

## SKILLS & TOOLS

- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers.
- **Blockchain & Web3:** Solidity, Ethereum, Hyperledger; smart contract development for decentralised security systems.
- **Programming & Data:** Python, C/C++, MATLAB, JavaScript, SQL; Pandas, NumPy, Tableau, Power BI.
- **Telecom & Network Tools:** GSM, UMTS, LTE, IMS, Core/RAN/Transport; Huawei U2000, Ericsson OSS, Nokia OSS.
- **Research & HPC:** Weights & Biases, SLURM (HPC cluster scheduling), Overleaf, Git.

## REFERENCES


**Prof. Yong Xiang**
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Deakin University
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